

MIT-BIH Arrhythmia Database

Analyzing Abnormal
Heartbeats



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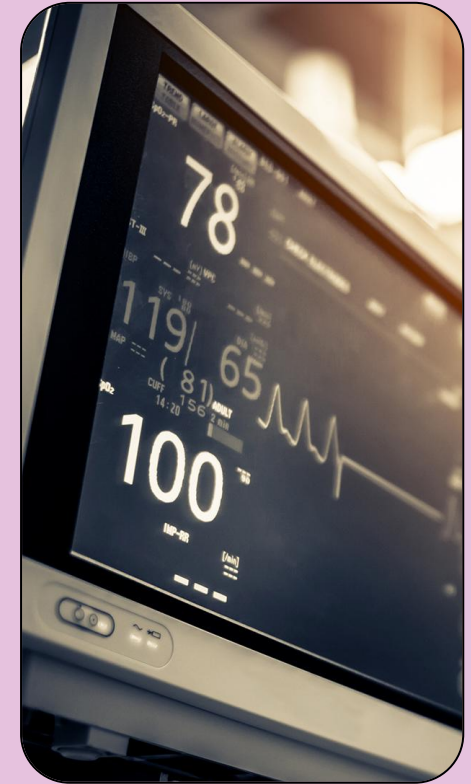
INTRODUCTION

- MIT-BIH Arrhythmia-Database through PhysioNet (1999)
- Boston's Beth Israel Hospital in collaboration with MIT
- Data includes 48 half-hour two-channel ECG recordings
 - 47 individuals (1975-1979)
 - Sampled at 360 Hz
 - 2 cardiologists
 - 110,000 heartbeats total
 - No missing data!
 - Research-grade
- Looking at ventricular beats:
 - **Transitions** between **normal** and **ventricular** (abnormal) beats
 - **Variability** in **timing** of the heartbeats



The Data

- We chose record 100 and 208
 - Record 100 = 2239 normal + 1 V beats + 33 A beats
 - Record 208 = 1586 normal + 992 V beats + 0 A beats
- Data includes 4 file types
 - .dat: Binary-encoded ECG waveforms
 - Record 100 = MLII, V5
 - Record 208 = MLII, V1
 - .atr: timing and type of heartbeat with symbolic codes
 - .hea: text file to explain heartbeat with channeling information
 - .xws: auxiliary file that contains waveform display settings



Data Extraction

Exploratory Data Analysis

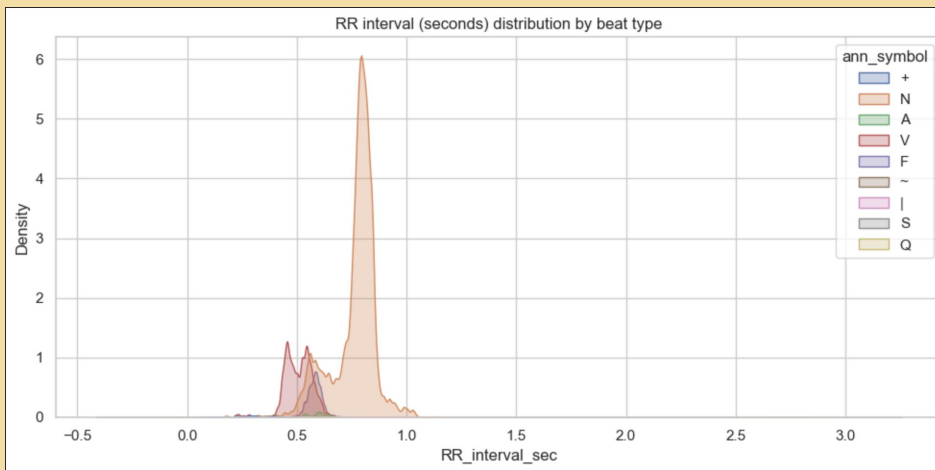
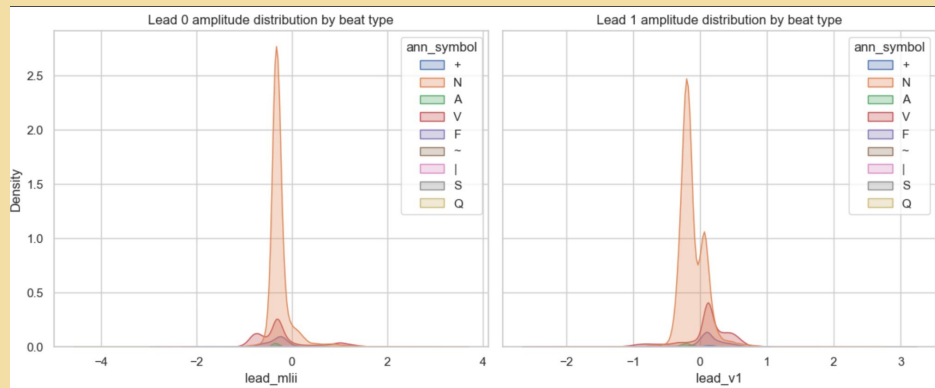
EDA

1. Class Imbalance

- Our dataset contains far more normal heartbeats (N) than ventricular beats (V)
- This reflects reality: most heartbeats are normal, even in patients with arrhythmias
- Challenge: We'll need to account for this imbalance when training our model to avoid it simply predicting "normal" for everything

2. Timing Differences

- Normal beats follow a regular rhythm with RR intervals around 0.8-1.0 seconds (equivalent to 60-75 beats per minute)
- PVCs arrive early: their RR intervals cluster around 0.4-0.6 seconds
 - a. PVCs are "premature" contractions that disrupt the normal rhythm
- The RR interval = a powerful predictor
 - a. if a beat arrives much earlier than expected, it's likely a PVC



Method 1: Random Forests

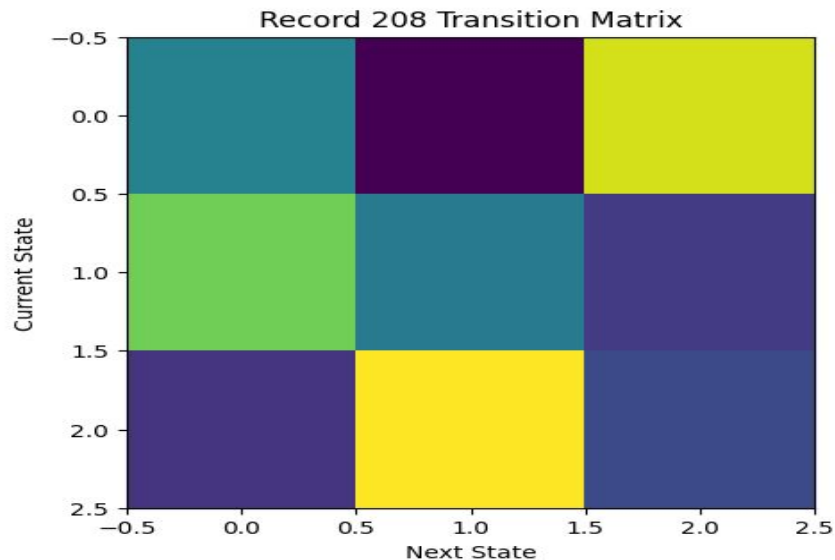
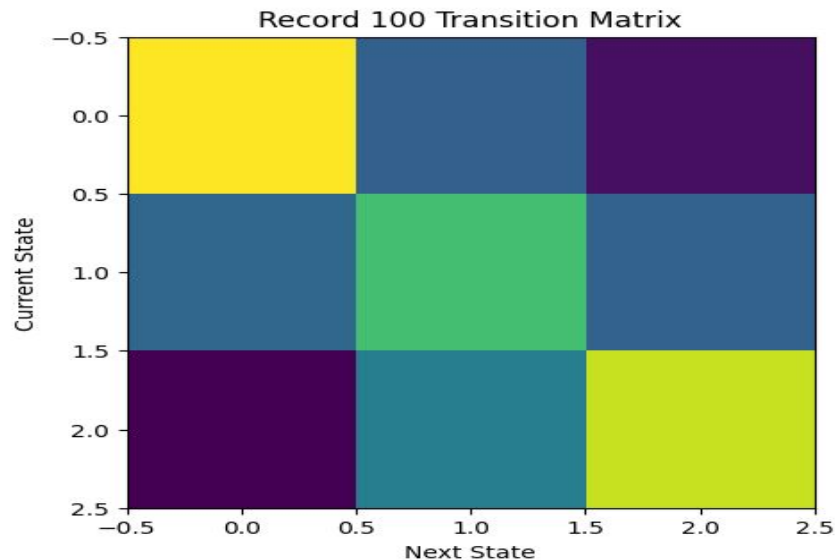
- **Our Goal:** Analyze and predict a PVC from a regular heartbeat.
- **Our Model** → Random Forest Classifier:
 - **Why:**
 - Captures Nonlinear patterns without assumptions
 - Handles noise and class imbalance
 - High Accuracy
 - **Alternatives:** Logistic Regression, KNN, CNN
 - We trained a Random Forest model using both timing (RR intervals) and shape (ECG amplitude) features to distinguish normal beats from PVCs.



Method 2: Markov Chain

Transition Matrices

Markov transition models: Capturing how the heart moves between different states (normal $\rightarrow$ PVC $\rightarrow$ back to normal)

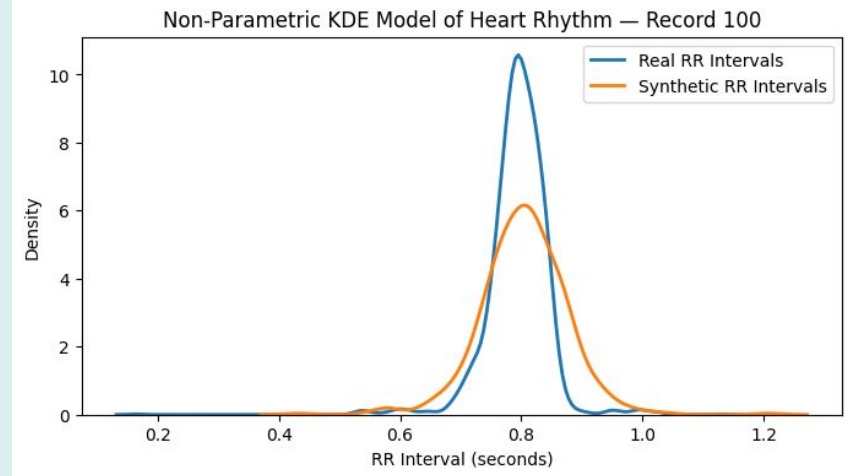
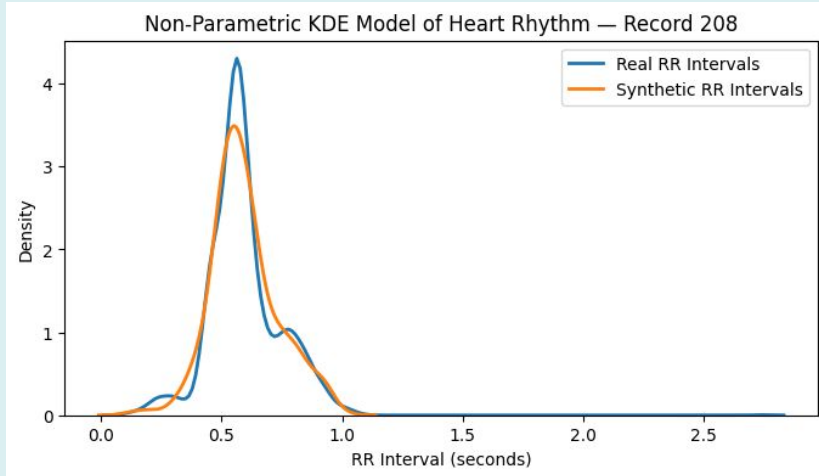


Method 3: KDE

KDE Plot

KDE: plots the PDF and show where RR interval values are concentrated, for 500 synthetic samples.

Results: Synthetic and Actual RR Intervals follow same shape with strong peaks at 0.5 – 0.6 (100 –120 BPM) and 0.8 (75 BPM) → steady and natural heartbeat

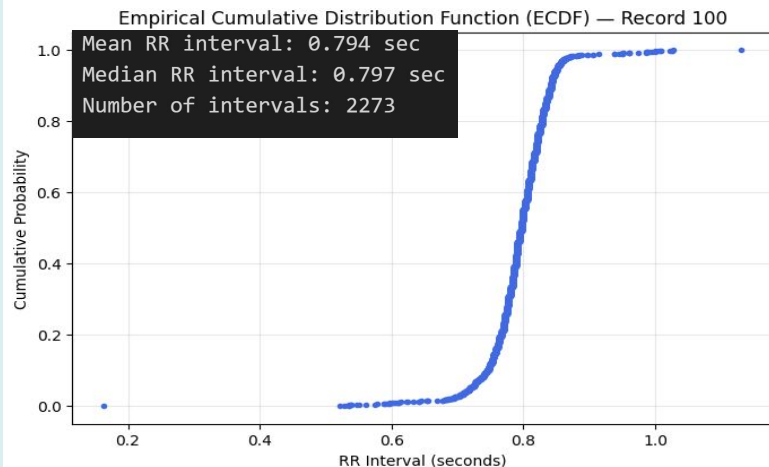
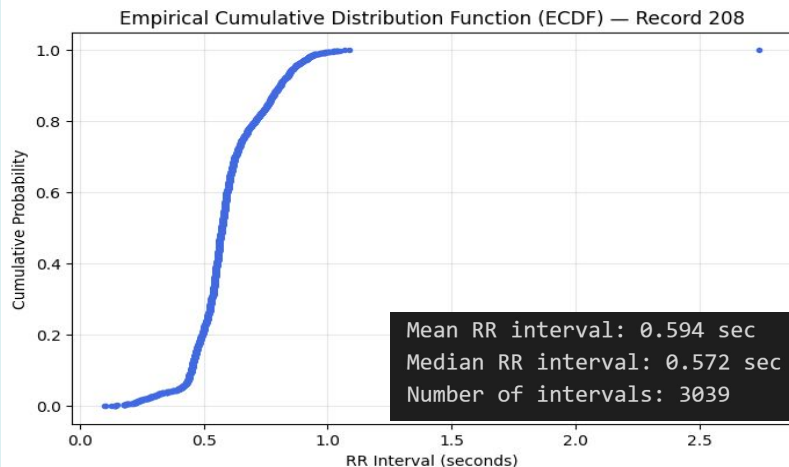


Method 4: ECDF

ECDF Plot

ECDF: tells us the proportion of observations less than or equal to a given data value.

Results: Both ECDF curves are steep, indicating tight range → heart beats are consistent and steady



Critical Evaluation

- Synthetic RR sequences vs Real RR sequences
- Bootstrapping did NOT capture underlying issues
 - Temporal dependencies
 - Rhythm patterns

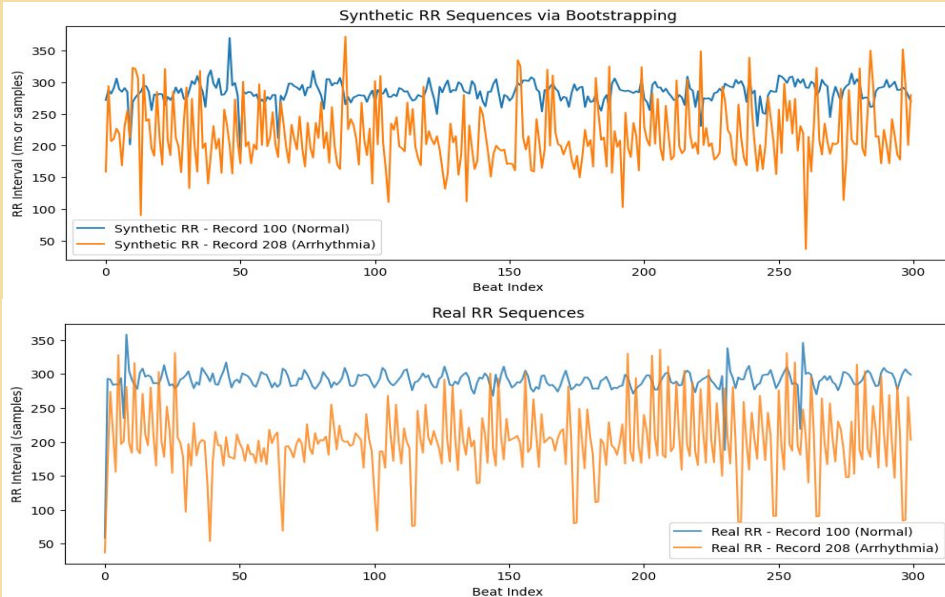
Differences:

- Overlap
- Irregularity
- Less separation

Uncertainty:

- Uncertainty is present
- Less representative and robust

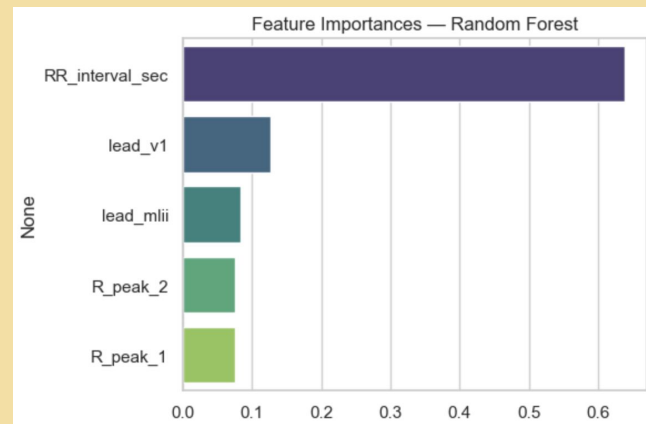
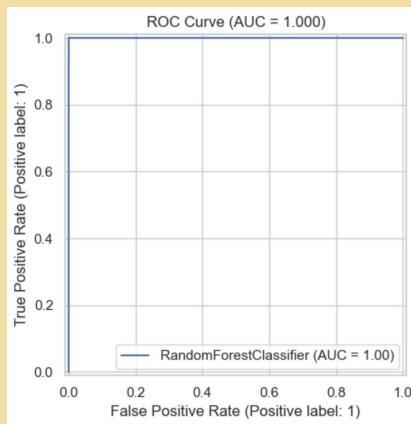
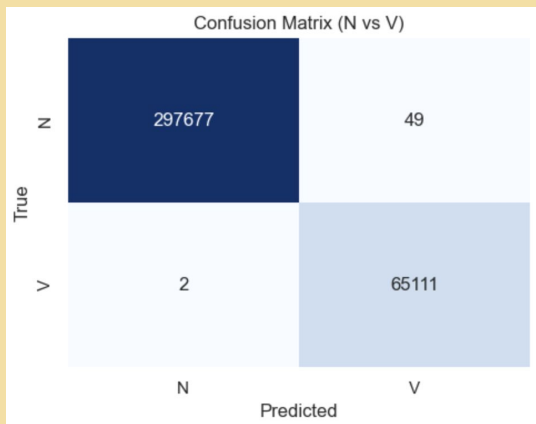
Results



Results

Modeling Process

- **Feature Importance:**
 - **RR_interval_sec** (Timing): ***By far the most important feature*** (~60% importance)
 - Confirms our EDA finding: PVCs arrive early, and this timing signature is the strongest predictor
 - **lead_v1** (Shape from ECG Lead 1): Second most important (~12% importance)
 - The electrical signal shape also helps distinguish PVCs
 - **lead_mlii** (Shape from Modified Lead II): Third (~10% importance)
 - **R_peak amplitudes:** Contribute modestly to prediction
- **Conclusion:** Our model can reliably detect PVCs in ECG data with near-perfect accuracy.



Results

- Normal rhythm (Record 100): Synthetic closely matched natural.
- Arrhythmic rhythm (Record 208): Synthetic didn't match natural.

Limitations

- The first-order Markov model only considers one previous beat, missing longer-term patterns.
- Biggest challenge for KDE and ECDF is choosing the sample size – if too small not representative of true shape and if too large distribution is too smooth

Future Work

- Use a higher-order Markov model so predictions depend on multiple prior beats.
- Adding hidden states to represent physiological conditions
- KDE and ECDF match underlying distribution, meaning there is no need to adjust